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Vanity mirror with hidden sensor

Abstract

The present disclosure relates to a mirror assembly having a housing, a first mirror, a second mirror, a light source, and a sensor assembly. The first mirror can be positioned within an opening of the housing. The front surface of the second mirror can be coupled to a rear surface of the first mirror. The light source can be disposed on or at least partially around the first mirror. The sensor assembly can be positioned between a rear surface of the second mirror and the housing. The sensor assembly can include a transmitter positioned behind the rear surface of the second mirror and a receiver positioned behind the rear surface of the second mirror. The light source can be activated when the sensor assembly detects a user's presence.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
D44537	12/1912	Mclsaac	N/A	N/A
D51556	12/1916	Hawthorne	N/A	N/A
1338582	12/1919	Morris et al.	N/A	N/A
1451236	12/1922	Stanfiled	N/A	N/A
D65759	12/1923	Short	N/A	N/A
1541451	12/1924	Wallace	N/A	N/A
2004166	12/1934	Low	N/A	N/A
2201251	12/1939	Patten	N/A	N/A
2235281	12/1940	Carver	N/A	N/A
2292059	12/1941	Charles	N/A	N/A
D163481	12/1950	Rauh	N/A	N/A
2687674	12/1953	Emilea	N/A	N/A
2737852	12/1955	Porter et al.	N/A	N/A
3006252	12/1960	Kacowski	N/A	N/A
3045551	12/1961	Bonanno	N/A	N/A
3315932	12/1966	Chandler	N/A	N/A
D208234	12/1966	Ely	N/A	N/A
D209077	12/1966	Andre	N/A	N/A
D213392	12/1968	Andre	N/A	N/A
D216414	12/1968	Hanson	N/A	N/A
3575496	12/1970	Pollock et al.	N/A	N/A
3623356	12/1970	Bisberg	N/A	N/A
3711179	12/1972	Takeda	N/A	N/A
3732702	12/1972	Desch	N/A	N/A
3751140	12/1972	Berlin et al.	N/A	N/A
3751141	12/1972	Brown	N/A	N/A
3794828	12/1973	Arpino	N/A	N/A
3825324	12/1973	Brewington	N/A	N/A
3914029	12/1974	Hoplock	N/A	N/A
3949767	12/1975	Rose	N/A	N/A
D243301	12/1976	Ravn	N/A	N/A
D243478	12/1976	Jones	N/A	N/A
4129355	12/1977	Noguchi	N/A	N/A
D254208	12/1979	Breslow	N/A	N/A
4278870	12/1980	Carleton et al.	N/A	N/A
D261845	12/1980	Wachtel	N/A	N/A
D266028	12/1981	Boyd	N/A	N/A
4491899	12/1984	Fleming	N/A	N/A
D284483	12/1985	Yang	N/A	N/A
D290662	12/1986	Basil et al.	N/A	N/A
D307358	12/1989	Gerton	N/A	N/A
D309833	12/1989	Wahl	N/A	N/A
D317531	12/1990	Evans	N/A	N/A
5025354	12/1990	Kondo	N/A	N/A
5093748	12/1991	Higdon	N/A	N/A
5164861	12/1991	Katz	N/A	N/A
D335580	12/1992	Gaullier	N/A	N/A
5217296	12/1992	Tanner et al.	N/A	N/A
5267081	12/1992	Pein	N/A	N/A
5267786	12/1992	Aisley	N/A	N/A

5392162	12/1994	Glucksman	N/A	N/A
D378159	12/1996	Mulkey	N/A	N/A
D379125	12/1996	Simjian	N/A	N/A
D391773	12/1997	Zaidman et al.	N/A	N/A
5796176	12/1997	Kramer et al.	N/A	N/A
D409003	12/1998	Scavini	N/A	N/A
5979976	12/1998	Ferencik	N/A	N/A
5984485	12/1998	Poli et al.	N/A	N/A
6042242	12/1999	Chang	N/A	N/A
D425313	12/1999	Zadro	N/A	N/A
D426182	12/1999	Brown	N/A	N/A
6106121	12/1999	Buckley et al.	N/A	N/A
D431375	12/1999	Zadro	N/A	N/A
6158877	12/1999	Zadro	N/A	N/A
6206530	12/2000	Eberts	N/A	N/A
D442371	12/2000	Eberts	N/A	N/A
6231992	12/2000	Niebauer et al.	N/A	N/A
6241357	12/2000	Lee	N/A	N/A
6270240	12/2000	Inoue	N/A	N/A
6273585	12/2000	Wu	N/A	N/A
6305809	12/2000	Zadro	N/A	N/A
D454701	12/2001	Eric	N/A	N/A
D459094	12/2001	Stone et al.	N/A	N/A
6420682	12/2001	Sellgren et al.	N/A	N/A
6466826	12/2001	Nishihira et al.	N/A	N/A
D465490	12/2001	Wei	N/A	N/A
6496107	12/2001	Himmelstein	N/A	N/A
6553123	12/2002	Dykstra	N/A	N/A
D474432	12/2002	Good	N/A	N/A
6560027	12/2002	Meine	N/A	N/A
6594630	12/2002	Zlokarnik et al.	N/A	N/A
6604836	12/2002	Carlucci et al.	N/A	N/A
6676272	12/2003	Chance	N/A	N/A
D486964	12/2003	Prince et al.	N/A	N/A
D488626	12/2003	Kruger	N/A	N/A
D492230	12/2003	Berger	N/A	N/A
6830154	12/2003	Zadro	N/A	N/A
6848822	12/2004	Ballen et al.	N/A	N/A
6854852	12/2004	Zadro	N/A	N/A
D505555	12/2004	Snell	N/A	N/A
6886351	12/2004	Palfy et al.	N/A	N/A
D508883	12/2004	Falconer	N/A	N/A
D509369	12/2004	Snell	N/A	N/A
D511413	12/2004	Yue	N/A	N/A
6961168	12/2004	Agrawal et al.	N/A	N/A
D512841	12/2004	Dirks	N/A	N/A
7004599	12/2005	Mullani	N/A	N/A
7048406	12/2005	Shih	362/135	A45D 42/10
7054668	12/2005	Endo et al.	N/A	N/A
D524469	12/2005	Pitot et al.	N/A	N/A
7090378	12/2005	Zadro	N/A	N/A
D532981	12/2005	Zadro	N/A	N/A
D540549	12/2006	Yue	N/A	N/A
7233154	12/2006	Groover et al.	N/A	N/A
D546567	12/2006	Bhavnani	N/A	N/A
D547555	12/2006	Lo et al.	N/A	N/A
D558987	12/2007	Gildersleeve	N/A	N/A
D562571	12/2007	Pitot	N/A	N/A

7341356	12/2007	Zadro	N/A	N/A
7347573	12/2007	Isler	N/A	N/A
7349144	12/2007	Varaprasad et al.	N/A	N/A
D568081	12/2007	Thompson et al.	N/A	N/A
D569671	12/2007	Thompson et al.	N/A	N/A
7370982	12/2007	Bauer et al.	N/A	N/A
D572024	12/2007	Shapiro	N/A	N/A
7393115	12/2007	Tokushita et al.	N/A	N/A
D574159	12/2007	Howard	N/A	N/A
7417699	12/2007	Yun et al.	N/A	N/A
7423522	12/2007	O'brien et al.	N/A	N/A
7435928	12/2007	Platz	N/A	N/A
7446924	12/2007	Schofield et al.	N/A	N/A
7455412	12/2007	Rottcher	N/A	N/A
D582984	12/2007	Mininger et al.	N/A	N/A
D584516	12/2008	Otomo	N/A	N/A
7500755	12/2008	Ishizaki et al.	N/A	N/A
7513476	12/2008	Huang	N/A	N/A
7551354	12/2008	Horsten et al.	N/A	N/A
7570413	12/2008	Tonar et al.	N/A	N/A
7589893	12/2008	Rottcher	N/A	N/A
7621651	12/2008	Chan et al.	N/A	N/A
7626655	12/2008	Yamazaki et al.	N/A	N/A
7636195	12/2008	Nieuwkerk et al.	N/A	N/A
7651229	12/2009	Rimback et al.	N/A	N/A
7679809	12/2009	Tonar et al.	N/A	N/A
7728927	12/2009	Nieuwkerk et al.	N/A	N/A
7805260	12/2009	Mischel, Jr.	362/135	G09F 9/00
D625930	12/2009	Merica	N/A	N/A
7813023	12/2009	Baur	N/A	N/A
7813060	12/2009	Bright et al.	N/A	N/A
7826123	12/2009	McCabe et al.	N/A	N/A
7853414	12/2009	Mischel, Jr. et al.	N/A	N/A
7855755	12/2009	Weller et al.	N/A	N/A
7856248	12/2009	Fujisaki	N/A	N/A
7859737	12/2009	McCabe et al.	N/A	N/A
7859738	12/2009	Baur et al.	N/A	N/A
7864399	12/2010	McCabe et al.	N/A	N/A
D635009	12/2010	Paterson	N/A	N/A
7898719	12/2010	Schofield et al.	N/A	N/A
7903335	12/2010	Nieuwkerk et al.	N/A	N/A
7916129	12/2010	Lin et al.	N/A	N/A
7916380	12/2010	Tonar et al.	N/A	N/A
7953648	12/2010	Vock	N/A	N/A
D639077	12/2010	DeBretton Gordon	N/A	N/A
7978393	12/2010	Tonar et al.	N/A	N/A
8004741	12/2010	Tonar et al.	N/A	N/A
D647444	12/2010	Manukyan et al.	N/A	N/A
D649790	12/2010	Pitot	N/A	N/A
8083386	12/2010	Lynam	N/A	N/A
D652220	12/2011	Pitot	N/A	N/A
8099247	12/2011	Mischel, Jr. et al.	N/A	N/A
D656979	12/2011	Yip et al.	N/A	N/A
D657425	12/2011	Podd	N/A	N/A
D657576	12/2011	Pitot	N/A	N/A
8154418	12/2011	Peterson et al.	N/A	N/A
8162502	12/2011	Zadro	N/A	N/A
D658604	12/2011	Egawa et al.	N/A	N/A

D660367	12/2011	Podd	N/A	N/A
D660368	12/2011	Podd	N/A	N/A
D660369	12/2011	Podd	N/A	N/A
8179236	12/2011	Weller et al.	N/A	N/A
8179586	12/2011	Schofield et al.	N/A	N/A
8194133	12/2011	DeWind et al.	N/A	N/A
8228588	12/2011	McCabe et al.	N/A	N/A
D665030	12/2011	Podd	N/A	N/A
D666010	12/2011	Farley	N/A	N/A
D670087	12/2011	Walker	N/A	N/A
8335032	12/2011	McCabe et al.	N/A	N/A
8348441	12/2012	Skelton	N/A	N/A
8356908	12/2012	Zadro	N/A	N/A
8379289	12/2012	Schofield et al.	N/A	N/A
8382189	12/2012	Li et al.	N/A	N/A
8393749	12/2012	Daicos	N/A	N/A
8400704	12/2012	McCabe et al.	N/A	N/A
D679101	12/2012	Pitot	N/A	N/A
D679102	12/2012	Gilboe et al.	N/A	N/A
D680755	12/2012	Gilboe et al.	N/A	N/A
8503062	12/2012	Baur et al.	N/A	N/A
8506096	12/2012	McCabe et al.	N/A	N/A
8508832	12/2012	Baumann et al.	N/A	N/A
8511841	12/2012	Varaprasad et al.	N/A	N/A
D688883	12/2012	Gilboe et al.	N/A	N/A
D689701	12/2012	Mischel, Jr. et al.	N/A	N/A
8559092	12/2012	Bugno et al.	N/A	N/A
8559093	12/2012	Varaprasad et al.	N/A	N/A
8585273	12/2012	Pokrovskiy et al.	N/A	N/A
D699448	12/2013	Yang et al.	N/A	N/A
D699952	12/2013	Yang et al.	N/A	N/A
8649082	12/2013	Baur	N/A	N/A
D701050	12/2013	Yang et al.	N/A	N/A
D701507	12/2013	Cope	N/A	N/A
8705161	12/2013	Schofield et al.	N/A	N/A
8727547	12/2013	McCabe et al.	N/A	N/A
D707454	12/2013	Pitot	N/A	N/A
8743051	12/2013	Moy et al.	N/A	N/A
D711871	12/2013	Daniel	N/A	N/A
D711874	12/2013	Cope	N/A	N/A
8797627	12/2013	McCabe et al.	N/A	N/A
D712963	12/2013	Fleet	N/A	N/A
8880360	12/2013	Mischel, Jr. et al.	N/A	N/A
8910402	12/2013	Mischel, Jr. et al.	N/A	N/A
D727630	12/2014	Zadro	N/A	N/A
D729525	12/2014	Tsai	N/A	N/A
D729527	12/2014	Tsai	N/A	N/A
D730065	12/2014	Tsai	N/A	N/A
9068737	12/2014	Kirchberger et al.	N/A	N/A
9090211	12/2014	McCabe et al.	N/A	N/A
D736001	12/2014	Yang et al.	N/A	N/A
D737059	12/2014	Tsai	N/A	N/A
D737060	12/2014	Yang et al.	N/A	N/A
9105202	12/2014	Mischel, Jr. et al.	N/A	N/A
D737580	12/2014	Tsai	N/A	N/A
D738118	12/2014	Gyanendra et al.	N/A	N/A
9170353	12/2014	Chang	N/A	N/A
9173509	12/2014	Mischel, Jr. et al.	N/A	N/A

9174578	12/2014	Uken et al.	N/A	N/A
9205780	12/2014	Habibi et al.	N/A	N/A
9232846	12/2015	Fung	N/A	N/A
9254789	12/2015	Anderson et al.	N/A	N/A
D751829	12/2015	Yang et al.	N/A	N/A
D754446	12/2015	Yang et al.	N/A	N/A
9327649	12/2015	Habibi	N/A	N/A
9341914	12/2015	McCabe et al.	N/A	N/A
9347660	12/2015	Zadro	N/A	N/A
D764592	12/2015	Zenoff	N/A	N/A
9499103	12/2015	Han	N/A	N/A
9510711	12/2015	Tsibulevskiy et al.	N/A	N/A
9528695	12/2015	Adachi et al.	N/A	N/A
D776945	12/2016	Yang	N/A	N/A
D779836	12/2016	Bailey	N/A	N/A
D785345	12/2016	Yang et al.	N/A	N/A
9638410	12/2016	Yang et al.	N/A	N/A
9694751	12/2016	Lundy, Jr. et al.	N/A	N/A
9709869	12/2016	Baumann et al.	N/A	N/A
D793099	12/2016	Bailey	N/A	N/A
9765958	12/2016	Lumaye et al.	N/A	N/A
D801060	12/2016	Hollinger	N/A	N/A
9827912	12/2016	Olesen et al.	N/A	N/A
9845537	12/2016	Mischel, Jr. et al.	N/A	N/A
9878670	12/2017	McCabe et al.	N/A	N/A
9897306	12/2017	Yang et al.	N/A	N/A
9921390	12/2017	Mischel, Jr. et al.	N/A	N/A
9933595	12/2017	Mischel, Jr. et al.	N/A	N/A
D816350	12/2017	Yang et al.	N/A	N/A
10016045	12/2017	Hollinger	N/A	N/A
10023123	12/2017	Takada et al.	N/A	N/A
10029616	12/2017	McCabe et al.	N/A	N/A
10035461	12/2017	Lin et al.	N/A	N/A
D825940	12/2017	Liu	N/A	N/A
10076176	12/2017	Yang et al.	N/A	N/A
D830706	12/2017	Pitot	N/A	N/A
10161622	12/2017	Frazier	N/A	N/A
D840699	12/2018	Xie	N/A	N/A
D845652	12/2018	Yang et al.	N/A	N/A
D846288	12/2018	Yang et al.	N/A	N/A
D848158	12/2018	Yang et al.	N/A	N/A
D850125	12/2018	Wang et al.	N/A	N/A
D854838	12/2018	Jeon et al.	N/A	N/A
D869863	12/2018	Liu	N/A	N/A
D871084	12/2018	Pestl et al.	N/A	N/A
10524591	12/2019	Kim	N/A	N/A
D874161	12/2019	Yang et al.	N/A	N/A
D874162	12/2019	Greenwalt	N/A	N/A
D878776	12/2019	Liu	N/A	N/A
D879481	12/2019	Yang	N/A	N/A
D879482	12/2019	Yang	N/A	N/A
D882280	12/2019	Yang	N/A	N/A
10652447	12/2019	Pestl et al.	N/A	N/A
D885769	12/2019	Wang	N/A	N/A
D891121	12/2019	Zhao et al.	N/A	N/A
D891123	12/2019	Li et al.	N/A	N/A
D891125	12/2019	Liu	N/A	N/A
10702043	12/2019	Yang et al.	N/A	N/A

D891792	12/2019	Yang	N/A	N/A
D892508	12/2019	Yang	N/A	N/A
D893892	12/2019	Rajasekaran et al.	N/A	N/A
10746394	12/2019	Yang et al.	N/A	N/A
D894615	12/2019	Yang	N/A	N/A
D895305	12/2019	Yang	N/A	N/A
D897694	12/2019	Lin	N/A	N/A
D897695	12/2019	Yang	N/A	N/A
D898383	12/2019	Zhou	N/A	N/A
D898386	12/2019	Huang	N/A	N/A
D898387	12/2019	Yang	N/A	N/A
D901908	12/2019	Zhao et al.	N/A	N/A
10869537	12/2019	Yang et al.	N/A	N/A
D909770	12/2020	Li et al.	N/A	N/A
D918602	12/2020	Chen	N/A	N/A
D919981	12/2020	Li	N/A	N/A
D919983	12/2020	Yang	N/A	N/A
D919984	12/2020	Yang	N/A	N/A
11013307	12/2020	Yang et al.	N/A	N/A
11026497	12/2020	Yang et al.	N/A	N/A
D924456	12/2020	Zhao et al.	N/A	N/A
D925928	12/2020	Yang et al.	N/A	N/A
D927863	12/2020	Yang et al.	N/A	N/A
D932198	12/2020	Sze	N/A	N/A
D932781	12/2020	Liu	N/A	N/A
D933374	12/2020	Luo	N/A	N/A
D935791	12/2020	Li et al.	N/A	N/A
D938170	12/2020	Liu	N/A	N/A
D939842	12/2021	Li et al.	N/A	N/A
D939843	12/2021	Li et al.	N/A	N/A
D942159	12/2021	Yang	N/A	N/A
D949579	12/2021	Chen	N/A	N/A
11371692	12/2021	Yang et al.	N/A	N/A
11549680	12/2022	Feit	N/A	F21V 23/005
11566784	12/2022	Yang et al.	N/A	N/A
11622614	12/2022	Yang et al.	N/A	N/A
11640042	12/2022	Yang	362/135	F21V 3/02
D990174	12/2022	Yang et al.	N/A	N/A
11708031	12/2022	Yang et al.	N/A	N/A
D1009485	12/2023	Yang et al.	N/A	N/A
2002/0196333	12/2001	Gorischek	N/A	N/A
2003/0030063	12/2002	Sosniak et al.	N/A	N/A
2003/0031010	12/2002	Sosniak et al.	N/A	N/A
2003/0065515	12/2002	Yokota	N/A	N/A
2003/0133292	12/2002	Mueller et al.	N/A	N/A
2003/0223250	12/2002	Ballen et al.	N/A	N/A
2004/0020509	12/2003	Waisman	N/A	N/A
2004/0027695	12/2003	Lin	N/A	N/A
2004/0125592	12/2003	Nagakubo et al.	N/A	N/A
2004/0156133	12/2003	Vernon	N/A	N/A
2004/0173498	12/2003	Lee	N/A	N/A
2005/0036300	12/2004	Dowling et al.	N/A	N/A
2005/0068646	12/2004	Lev et al.	N/A	N/A
2005/0146863	12/2004	Mullani	N/A	N/A
2005/0156753	12/2004	Deline et al.	N/A	N/A
2005/0243556	12/2004	Lynch	N/A	N/A
2005/0270769	12/2004	Smith	N/A	N/A
2005/0276053	12/2004	Nortrup et al.	N/A	N/A

2006/0006988	12/2005	Harter, Jr. et al.	N/A	N/A
2006/0077654	12/2005	Krieger et al.	N/A	N/A
2006/0132923	12/2005	Hsiao et al.	N/A	N/A
2006/0164725	12/2005	Horsten et al.	N/A	N/A
2006/0184993	12/2005	Goldthwaite et al.	N/A	N/A
2006/0186314	12/2005	Leung	N/A	N/A
2006/0286396	12/2005	Jonza	N/A	N/A
2007/0040033	12/2006	Rosenberg	N/A	N/A
2007/0050211	12/2006	Mandl	N/A	N/A
2007/0097672	12/2006	Benn	N/A	N/A
2007/0146616	12/2006	Nouchi et al.	N/A	N/A
2007/0159846	12/2006	Nishiyama et al.	N/A	N/A
2007/0183037	12/2006	De Boer et al.	N/A	N/A
2007/0263999	12/2006	Keam	N/A	N/A
2007/0297189	12/2006	Wu et al.	N/A	N/A
2008/0024864	12/2007	Alberti	N/A	N/A
2008/0078796	12/2007	Parsons	N/A	N/A
2008/0088244	12/2007	Morishita	N/A	N/A
2008/0118080	12/2007	Gratke et al.	N/A	N/A
2008/0130305	12/2007	Wang et al.	N/A	N/A
2008/0244940	12/2007	Mesika	N/A	N/A
2008/0258110	12/2007	Oshio	N/A	N/A
2008/0265799	12/2007	Sibert	N/A	N/A
2008/0271354	12/2007	Bostrom	N/A	N/A
2008/0294012	12/2007	Kurtz et al.	N/A	N/A
2008/0297586	12/2007	Kurtz et al.	N/A	N/A
2008/0298080	12/2007	Wu et al.	N/A	N/A
2009/0027902	12/2008	Fielding et al.	N/A	N/A
2009/0097252	12/2008	Liou et al.	N/A	N/A
2009/0194670	12/2008	Rains, Jr. et al.	N/A	N/A
2009/0207339	12/2008	Ajichi et al.	N/A	N/A
2009/0213604	12/2008	Uken	N/A	N/A
2009/0244740	12/2008	Takayanagi et al.	N/A	N/A
2009/0301927	12/2008	Fvlbrook et al.	N/A	N/A
2010/0033988	12/2009	Chiu et al.	N/A	N/A
2010/0118422	12/2009	Holacka	N/A	N/A
2010/0118520	12/2009	Stern et al.	N/A	N/A
2010/0214662	12/2009	Takayanagi et al.	N/A	N/A
2010/0296298	12/2009	Martin, Jr.	N/A	N/A
2010/0309159	12/2009	Roettcher	N/A	N/A
2011/0058269	12/2010	Su	N/A	N/A
2011/0074225	12/2010	Delnoij et al.	N/A	N/A
2011/0080374	12/2010	Feng et al.	N/A	N/A
2011/0168687	12/2010	Door et al.	N/A	N/A
2011/0194200	12/2010	Greenlee	N/A	N/A
2011/0211079	12/2010	Rolston	N/A	N/A
2011/0273659	12/2010	Sobecki	N/A	N/A
2011/0283577	12/2010	Cornelissen et al.	N/A	N/A
2012/0056738	12/2011	Lynam	N/A	N/A
2012/0080903	12/2011	Li et al.	N/A	N/A
2012/0081915	12/2011	Foote et al.	N/A	N/A
2012/0229789	12/2011	Kang et al.	N/A	N/A
2012/0307490	12/2011	Ellis	N/A	N/A
2013/0026512	12/2012	Tsai	N/A	N/A
2013/0077292	12/2012	Zimmerman	N/A	N/A
2013/0120989	12/2012	Sun et al.	N/A	N/A
2013/0190845	12/2012	Liu et al.	N/A	N/A
2013/0235607	12/2012	Yang et al.	N/A	N/A

2013/0235610	12/2012	Yang et al.	N/A	N/A
2014/0240964	12/2013	Adachi et al.	N/A	N/A
2014/0265768	12/2013	Diemel, Jr. et al.	N/A	N/A
2015/0060431	12/2014	Yang et al.	N/A	N/A
2015/0203970	12/2014	Mischel, Jr. et al.	N/A	N/A
2015/0205110	12/2014	Mischel, Jr. et al.	N/A	N/A
2015/0305113	12/2014	Ellis	N/A	N/A
2016/0045015	12/2015	Baldwin	N/A	N/A
2016/0070085	12/2015	Mischel, Jr. et al.	N/A	N/A
2016/0082890	12/2015	Habibi et al.	N/A	N/A
2016/0178964	12/2015	Sakai et al.	N/A	N/A
2016/0193902	12/2015	Hill et al.	N/A	N/A
2016/0200256	12/2015	Takada et al.	N/A	N/A
2016/0243989	12/2015	Habibi	N/A	N/A
2016/0255941	12/2015	Yang et al.	N/A	N/A
2017/0028924	12/2016	Baur et al.	N/A	N/A
2017/0139302	12/2016	Tonar	N/A	N/A
2017/0158139	12/2016	Tonar et al.	N/A	N/A
2017/0164719	12/2016	Wheeler	N/A	N/A
2017/0181541	12/2016	Stanley, Jr. et al.	N/A	N/A
2017/0188020	12/2016	Sakai et al.	N/A	N/A
2017/0190290	12/2016	Lin et al.	N/A	N/A
2017/0257543	12/2016	Rowles et al.	N/A	N/A
2017/0285392	12/2016	Hirata et al.	N/A	N/A
2017/0297495	12/2016	Lundy, Jr. et al.	N/A	N/A
2017/0297498	12/2016	Larson et al.	N/A	N/A
2017/0313251	12/2016	Uken et al.	N/A	N/A
2017/0322389	12/2016	Hagestad et al.	N/A	N/A
2017/0349102	12/2016	Habibi	N/A	N/A
2018/0012526	12/2017	Dunn et al.	N/A	N/A
2018/0015880	12/2017	Olesen et al.	N/A	N/A
2018/0017823	12/2017	Saenger Nayver et al.	N/A	N/A
2018/0032227	12/2017	Broxson	N/A	N/A
2018/0050641	12/2017	Lin et al.	N/A	N/A
2018/0105114	12/2017	Geerlings et al.	N/A	N/A
2018/0147993	12/2017	McCabe et al.	N/A	N/A
2018/0162269	12/2017	Bredeweg et al.	N/A	N/A
2018/0172265	12/2017	Yang et al.	N/A	N/A
2018/0251069	12/2017	LaCross et al.	N/A	N/A
2018/0263362	12/2017	Yang et al.	N/A	N/A
2018/0268747	12/2017	Braun	N/A	N/A
2018/0270410	12/2017	Lyle et al.	N/A	N/A
2018/0343418	12/2017	Van Ness	N/A	N/A
2019/0000219	12/2018	Yang et al.	N/A	N/A
2019/0003699	12/2018	Mondora	N/A	N/A
2019/0054863	12/2018	Roth	N/A	N/A
2019/0086890	12/2018	Bradley et al.	N/A	N/A
2019/0087788	12/2018	Murphy et al.	N/A	N/A
2019/0089550	12/2018	Rexach et al.	N/A	N/A
2019/0138704	12/2018	Shrivastava et al.	N/A	N/A
2019/0172464	12/2018	Bargetzi et al.	N/A	N/A
2019/0246772	12/2018	Yang et al.	N/A	N/A
2019/0250781	12/2018	Savitski	N/A	N/A
2019/0267825	12/2018	Chien	N/A	N/A
2019/0291647	12/2018	Yang et al.	N/A	N/A
2019/0328161	12/2018	Wei	N/A	N/A
2019/0351830	12/2018	Bosma et al.	N/A	N/A
2020/0008592	12/2019	Meyers et al.	N/A	N/A

2020/0085170	12/2019	Yang et al.	N/A	N/A
2020/0268127	12/2019	Yang et al.	N/A	N/A
2020/0278514	12/2019	Yang et al.	N/A	N/A
2020/0333934	12/2019	Pestl et al.	N/A	N/A
2021/0025584	12/2020	Yang et al.	N/A	N/A
2021/0137266	12/2020	Pestl et al.	N/A	N/A
2021/0196028	12/2020	Yang et al.	N/A	N/A
2021/0307491	12/2020	Yang et al.	N/A	N/A
2021/0364892	12/2020	Copeland et al.	N/A	N/A
2022/0282861	12/2021	Yang et al.	N/A	N/A
2023/0102011	12/2022	Yang et al.	N/A	N/A
2023/0204201	12/2022	Yang et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
3044427	12/1995	CN	N/A
1190201	12/1997	CN	N/A
2379638	12/1999	CN	N/A
3357935	12/2003	CN	N/A
2852806	12/2006	CN	N/A
2925206	12/2006	CN	N/A
300746709	12/2007	CN	N/A
101160003	12/2007	CN	N/A
101382025	12/2008	CN	N/A
300973066	12/2008	CN	N/A
300983799	12/2008	CN	N/A
300990023	12/2008	CN	N/A
301001894	12/2008	CN	N/A
201335322	12/2008	CN	N/A
201388790	12/2009	CN	N/A
301108997	12/2009	CN	N/A
301209880	12/2009	CN	N/A
101787830	12/2009	CN	N/A
301278203	12/2009	CN	N/A
301340032	12/2009	CN	N/A
301502988	12/2010	CN	N/A
102057756	12/2010	CN	N/A
102109129	12/2010	CN	N/A
301583101	12/2010	CN	N/A
301811715	12/2011	CN	N/A
302103915	12/2011	CN	N/A
302140631	12/2011	CN	N/A
302140632	12/2011	CN	N/A
302337970	12/2012	CN	N/A
302363850	12/2012	CN	N/A
302396166	12/2012	CN	N/A
302442518	12/2012	CN	N/A
103300590	12/2012	CN	N/A
302638575	12/2012	CN	N/A
302668773	12/2012	CN	N/A
204146556	12/2014	CN	N/A
205265762	12/2015	CN	N/A
205072328	12/2015	CN	N/A
106377049	12/2016	CN	N/A
206119539	12/2016	CN	N/A
106723885	12/2016	CN	N/A
206371658	12/2016	CN	N/A
108185741	12/2017	CN	N/A
108308888	12/2017	CN	N/A

207626762	12/2017	CN	N/A
108713949	12/2017	CN	N/A
211577476	12/2019	CN	N/A
211600392	12/2019	CN	N/A
111759073	12/2019	CN	N/A
306124222	12/2019	CN	N/A
308268382	12/2022	CN	N/A
308330983	12/2022	CN	N/A
308338019	12/2022	CN	N/A
2924529	12/1980	DE	N/A
29904039	12/1998	DE	N/A
20014279	12/2000	DE	N/A
102004042929	12/2005	DE	N/A
202007013393	12/2006	DE	N/A
102006060781	12/2007	DE	N/A
202009004795	12/2008	DE	N/A
202010000170	12/2009	DE	N/A
202012103555	12/2013	DE	N/A
0367134	12/1989	EP	N/A
1792553	12/2006	EP	N/A
2 788 951	12/1999	FR	N/A
2346206	12/1999	GB	N/A
2363712	12/2001	GB	N/A
S49-131097	12/1973	JP	N/A
55-129073	12/1979	JP	N/A
59-166769	12/1983	JP	N/A
S62-112931	12/1986	JP	N/A
H05-009413	12/1992	JP	N/A
3057292	12/1997	JP	N/A
2003-79495	12/2002	JP	N/A
2004-290531	12/2003	JP	N/A
2006-202602	12/2005	JP	N/A
2008-073174	12/2007	JP	N/A
2013-172802	12/2012	JP	N/A
2014-212075	12/2013	JP	N/A
2016-168171	12/2015	JP	N/A
2017-144039	12/2016	JP	N/A
2018-167022	12/2017	JP	N/A
7497367	12/2023	JP	N/A
30-0318286	12/2002	KR	N/A
2003-0017261	12/2002	KR	N/A
30-0330692	12/2002	KR	N/A
200400903	12/2004	KR	N/A
30-0507873	12/2007	KR	N/A
30-0586341	12/2010	KR	N/A
30-0692452	12/2012	KR	N/A
30-0712086	12/2012	KR	N/A
WO 2004/074886	12/2003	WO	N/A
WO 2013/047784	12/2012	WO	N/A
WO 2018/045649	12/2017	WO	N/A
WO 2020/061091	12/2019	WO	N/A

OTHER PUBLICATIONS

U.S. Appl. No. 29/631,301, filed Dec. 28, 2017, Yang et al. cited by applicant
U.S. Appl. No. 29/662,730, filed Sep. 7, 2018, Yang et al. cited by applicant
U.S. Appl. No. 29/689,860, filed May 2, 2019, Yang et al. cited by applicant
Ilumay M-97 Led Smart Sensor Mirror, available from internet https://www.alibaba.com/product-detail/ilumay-M-97-led-smart-sensor_60701769220.html, availablility as early as Dec. 16, 2017. cited by applicant

Advanced Lighting Guidelines, 1993 (second edition), Chapter entitled, “Occupant Sensors”, Published by California Energy Commission (CEC Pub.), in 14 pages. cited by applicant
Jerdon, Model JRT910CL 5X Magnified Lighted Tabletop Rectangular Mirror, Chrome Finish, 67.2 Ounce, https://www.amazon.com/Jerdon-JRT910CL-Magnified-Tabletop-Rectangular/dp/B00N1WE3UC?th=1, Jun. 2015, in 8 pages. cited by applicant
Kore, “Building an intelligent voice controlled mirror,” retrieved from the internet on Jul. 11, 2019: https://medium.com/@akshaykore/building-an-intelligent-voice-controlled-mirror-2edbc7d62c9e, Jun. 26, 2017, in 10 pages. cited by applicant
Pinterest, Plug-in wall-mount makeup mirror has adjustment handle, https://www.pinterest.com/pin/856035841641838288/?d=t&mt=login, viewed on Feb. 1, 2022, in 3 pages. cited by applicant
Sharper Image, Model JRT718CL Product Specification, Slimline Series LED Lighted Wall Mount Mirror, copyright 2015, https://www.ameraproducts.com/Shard/ProductLitirature/Jerdon/JRT718CL-amera.pdf, in 1 pages. cited by applicant
Sharper Image, Model JRT950NL , Slimline LED Lighted Tabletop 8X Magnification Mirror, https://www.amazon.com/Sharper-Imange-JRT950NL-Slimline-Magnification/dp/B015W76T3M?th=1, Jan. 20, 2016, in 8 pages. cited by applicant
Simple Human Vanity Mirror, available from internet at http://www.bedbathandbeyond.com/store/products/simplehuman-reg-5x-sensor-vanity- mirror/1041483503?categoryId=12028, apparently available Dec. 19, 2013, site visited Dec. 2, 2014. cited by applicant
Simple Human Sensor Mirror, Internet Archive Wayback Machine webpage capture of http://www.tuvie.com/stainless-steel-sensor-mirror-by-simplehuman/, apparently available Jan. 27, 2013, site visited Dec. 2, 2014. cited by applicant
Simplehuman Mini Sensor Mirror, available from internet at http://www.amazon.com/gp/product/B00FZ3MFAA/ref=pd_lpo_sbs_dp_ss_2?pf_rd_p=1944579862&pf_rd_s=lpo-top-stripe-1&pf_rd_t=201&pf_rd_i=B00M8MC5H4&pf_rd_m=ATVPDKIKXODER&pf_rd_r=0RHFJEABM9QKSWJKK99N#Ask apparently available Mar. 11, 2014, site visited Jan. 8, 2015. cited by applicant
Simplehuman Sensor Mirror, available from internet at http://www.amazon.com/simplehuman-Sensor-Sensor-Activated-Lighted-Magnification/dp/B00M8MC5H4#customerReviews, apparently available Dec. 31, 2014, site visited Jan. 8, 2015. cited by applicant
Simplehuman Wall Mount Mirror, available from internet at http://www.amazon.com/simplehuman-Wall-Mount-Sensor-Mirror/dp/B00FN92ELG#customerReviews, available at least as early as Jan. 31, 2013, site visited Jan. 8, 2015. cited by applicant
Simplehuman Wide View Sensor Mirror, available from internet at http://www.amazon.com/simplehuman-Wide-View-Sensor-Mirror/dp/B01C2RXD7K, site visited Aug. 9, 2016. cited by applicant
Simplehuman Sensor Mirror Pro Wide-View, available from internet at http://www.simplehuman.com/wide-view-sensor-mirror, site visited Aug. 9, 2016. cited by applicant
Brookstone Shower Mirror, available from internet at http://www.brookstone.com/9-Lighted-Fogless-Shower-Mirror?bkiid=? SubCategory_Bath_Spa_Mirrors_Lighting_Makeup_Mirrors%7CSubCategoryWidget%7C608364p&catld=n/, apparently available Jan. 15, 2013, site visited Dec. 2, 2014. cited by applicant
Jerdon Wall Mounted Mirror, available from internet at http://www.amazon.com/Jerdon-HL1016NL-9-5-Inch-Lighted-Magnification/dp/B00413G9K2/ref=sr_1_26?ie=UTF8&qid=1420579897&sr=8-26&keywords=wall+mounted+mirror#customerReviews, apparently available Feb. 21, 2009, site visited Jan. 8, 2015. cited by applicant
Jerdon Wall Mounted Mirror, available from internet at http://www.amazon.com/Jerdon-JD7C-9-Inch-Lighted-Magnification/dp/B001DKVC08/ref=sr_1_54?ie=UTF8&qid=1420580127&sr=8-54&keywords=wall+mounted+mirror, apparently available Oct. 6, 2010, site visited Jan. 8, 2015. cited by applicant
Zadro Z'fogless Mirror with Light, available from internet at http://www.amazon.com/Zadro-1X-Zfogless-Adjustable-Magnification/dp/B000ARWLIW/ref=sr_1_16?s=beauty&ie=UTF8&qid=1439229012&sr=1-16&keywords=zadro+lighted+fogless+mirror, apparently available Nov. 27, 2006, site visited Aug. 10, 2015. cited by applicant

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS (1) Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57. For example, this application claims the benefit of U.S. Patent Application No. 63/488,389 titled "VANITY MIRROR WITH HIDDEN SENSOR" and filed Mar. 3, 2023, the entirety of which is hereby incorporated by reference for all purposes and forms a part of this specification.

BACKGROUND

Field

(1) The present disclosure relates to reflective devices such as mirrors, in particular to mirrors having lights that are activated when a user's presence is detected.

Description of the Related Art

(2) Vanity mirrors are mirrors that are typically used for reflecting an image of a user during personal grooming, primping, cosmetic care, or the like. Vanity mirrors are available in different configurations, such as free-standing mirrors, hand-held mirrors, mirrors connected to vanity tables, bathroom wall mirrors, car mirrors, and/or mirrors attached to or produced by electronic screens or devices.

SUMMARY

(3) The following disclosure describes non-limiting examples of some embodiments. For instance, other embodiments according to the present disclosure may or may not include any particular feature or any particular set of features described herein. Moreover, disclosed advantages and benefits may apply only to certain embodiments of the invention and should not be used to limit the disclosure.

(4) In a mirror assembly that includes a proximity sensor for sensing the presence of a user and a controller for activating a user-illuminating light source when a user is sensed, the proximity sensor can be hidden from the user's view upon casual observation by the user. For example, the proximity sensor can be positioned behind or distal from the front, outer, or proximal-most reflective surface of the mirror assembly. The proximity sensor can be positioned on the opposite or back side of the outer or proximal-most reflective surface from the user. In some embodiments, the mirror assembly can hide the proximity sensor from the user by providing a front region of the mirror assembly that reflects a majority of visible light but that permits non-visible light or another form of sensing energy, such as acoustic waves, to pass through this region of the mirror assembly. The sensor positioned behind the front reflective surface can be configured to transmit sensing energy through a region of the mirror assembly toward the user and to receive the reflected energy back from the user, while permitting a majority of visible light that impinges on the front or outer surface of the region to be reflected back to the user.

(5) In some embodiments, a mirror assembly includes a housing, a first mirror, a second mirror, a light source, and a sensor assembly. The housing includes an opening. The first mirror is positioned within the opening of the housing. The first mirror has a front surface and a rear surface. The front surface is visible by a user. The second mirror has a front surface and a rear surface. Each of the first and second mirrors includes a thick substrate or layer of transparent material (e.g., glass or plastic). The front surface of the second mirror is coupled to the rear surface of the first mirror. The light source is disposed on or at least partially around the first mirror. The sensor assembly is positioned between a rear surface of the second mirror and the housing. The sensor assembly includes a transmitter positioned behind the rear surface of the second mirror and a receiver positioned behind the rear surface of the second mirror. The light source is activated when the sensor assembly detects a user's presence.

(6) In some embodiments, the sensor assembly includes a proximity sensor. In some embodiments, the transmitter and the receiver are positioned behind a window of the first mirror. In some embodiments, the mirror assembly includes an indicator or display configured to convey a signal or information to a user through a reflective surface of the mirror assembly, such as an LED light, positioned behind the window of the first mirror. In some embodiments, the LED light can be configured to emit one form of light (e.g., yellow light and/or flashing light) when the mirror assembly is charging, another form of light (e.g., green light and/or flashing light) when the mirror assembly is finished charging, and/or another form of light (e.g., red light and/or flashing light) when the mirror assembly has a low-power or near-depleted battery. In some embodiments, the window is substantially invisible to or hidden from the user upon casual observation in normal use. In some embodiments, the second mirror is smaller than the first mirror. In some embodiments, the first mirror and the second mirror have the same curvature.

(7) In some embodiments, the second mirror is coupled to the rear surface of the first mirror via an optical adhesive. In some embodiments, the optical adhesive is made of a material that mimics, or has substantially the same, index of refraction as the transparent material of the first mirror. In some embodiments, the first mirror has a layer or coating on the rear surface of the transparent material, and the second mirror has a layer or coating on the front surface of the transparent material. In some embodiments, the coating or layer of the first mirror comprises a reflective metal such as aluminum, and the layer or coating of the second mirror comprises a multilayer optical interference coating that includes a plurality of layers comprising two or more of layers comprising metal oxides

and/or dielectrics.
(8) In some embodiments, the mirror assembly includes a power source positioned within the housing. In some embodiments, the mirror assembly includes a shaft coupled at a first end to the housing and at a second end to a stand. In some embodiments, the shaft is rotatably coupled at the first end to the housing and the shaft is rotatably coupled at the second end to the stand.
(9) In some embodiments, a mirror assembly includes first mirror, a second mirror, and a sensor assembly. The first mirror has a front surface and a rear surface. The front surface is visible by a user. The second mirror has a front surface and a rear surface. The front surface of the second mirror is coupled to the rear surface of the first mirror. The sensor assembly is positioned behind a rear surface of the second mirror. The sensor assembly includes a transmitter configured to emit light and a receiver configured to receive light. The sensor assembly is configured to detect a user's presence.
(10) In some embodiments, the sensor assembly includes a proximity sensor. In some embodiments, the mirror assembly includes a light source, wherein when a user's presence is detected the light source is activated. In some embodiments, the transmitter and the receiver are positioned behind a window of the first mirror. In some embodiments, the second mirror is smaller than the first mirror. In some embodiments, the first mirror and the second mirror have the same curvature. In some embodiments, the second mirror is coupled to the rear surface of the first mirror via an adhesive. In some embodiments, the first mirror has a coating on the rear surface and the second mirror has a coating on the front surface. In some embodiments, the coating of the first mirror is aluminum and the coating of the second mirror is an optical interference coating.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Some of these drawings are schematic, showing some examples of basic parts and concepts. Many different or additional structures, implementations, components, mechanisms, steps, and processes can be used. The claimed inventions should not be limited in any way to anything illustrated in the drawings.
(2) FIG. **1** illustrates a front perspective view of an embodiment of a mirror assembly;
(3) FIG. **2** illustrates a front view of the mirror assembly of FIG. **1**;
(4) FIG. **3** illustrates a rear view of the mirror assembly of FIG. **1**;
(5) FIG. **4** illustrates a right side view of the mirror assembly of FIG. **1**;
(6) FIG. **5** illustrates a left side view of the mirror assembly of FIG. **1**;
(7) FIG. **6** illustrates a top view of the mirror assembly of FIG. **1**;
(8) FIG. **7** illustrates a bottom view of the mirror assembly of FIG. **1**;
(9) FIG. **8** illustrates a side cross-sectional view of the mirror assembly of FIG. **1**;
(10) FIG. **9** illustrates a sideview of the mirror assembly of FIG. **1** in a stowed configuration;
(11) FIG. **10** illustrates a top view of the mirror assembly of FIG. **1** in a stowed configuration;
(12) FIG. **11** illustrates a front view of the mirror assembly of FIG. **1** with a first mirror removed for illustrative purposes;
(13) FIG. **12** illustrates a side perspective view of the mirror assembly of FIG. **1** with a housing removed for illustrative purposes;
(14) FIG. **13** illustrates a front view of the mirror assembly of FIG. **1**;
(15) FIG. **14** illustrates a schematic cross-sectional side view of a portion of the layers of the first mirror and the second mirror of the mirror assembly of FIG. **1** when coupled;
(16) FIG. **15** illustrates a front view of the mirror assembly of FIG. **1** with the first mirror and the second mirror removed for illustrative purposes; and
(17) FIG. **16** illustrates a rear view of the mirror assembly of FIG. **1** with a back portion of the housing removed for illustrative purposes.
(18) FIG. **17** is a graph showing the relationship between reflectance and wavelength.
DETAILED DESCRIPTION
(19) This specification provides textual descriptions and illustrations of many devices, components, assemblies, and subassemblies. Any structure, material, function, method, or step that is described and/or illustrated in one example can be used by itself or with or instead of any structure, material, function, method, or step that is described and/or illustrated in another example or used in this field. The text and drawings merely provide examples and should not be interpreted as limiting or exclusive. No feature disclosed in this application is considered critical or indispensable. The relative sizes and proportions of the components illustrated in the drawings form part of the supporting disclosure of this specification but should not be considered to limit any claim unless recited in such claim.
(20) Certain embodiments of a mirror assembly are disclosed in the context of a portable, free-standing vanity

mirror, as it has particular utility in this context. However, the various aspects of the present disclosure can be used in many other contexts as well, such as wall-mounted mirrors, mirrors mounted on articles of furniture, automobile vanity mirrors (e.g., mirrors located in sun-visors), and otherwise. None of the features described herein are essentially or indispensable. Any feature, structure, or step disclosed herein can be replaced with or combined with any other feature, structure, or step disclosed herein, or omitted.

(21) Embodiments of the present disclosure relate to mirror assemblies. In particular, many embodiments of this disclosure relate to mirror assemblies having one or more light sources that are activated when a user's presence is detected. Some embodiments include one or more proximity sensors that are positioned behind a visual image reflective surface, such as a mirror. The positioning of the sensor(s) behind the mirror can provide an improved visual appearance of the mirror assembly because the sensor is hidden behind the mirror rather than visible on a user facing surface of the mirror assembly. In some implementations, this can provide the benefit of a simpler, more aesthetically pleasing appearance, and the benefit of permitting more light to be emitted more uniformly around the complete perimeter or circumference of the mirror, since the sensor does not occupy a portion of the light-emitting ring or periphery.

(22) FIGS. **1-8** illustrate an example embodiment of a mirror assembly **100**. The mirror assembly **100** can include a housing **104** and a visual image reflective surface, such as a mirror **108**. The mirror **108** can be positioned within an opening **112** of the housing **104**. In some embodiments, the mirror assembly **100** can include a shaft **116** that can couple the housing **104** to a stand **120** or base portion. The shaft **116** and/or the stand **120** can support the housing **104**. The housing **104** can comprise plastic, stainless steel, aluminum, or other suitable materials.

(23) In some embodiments the housing **104** can be pivotably coupled to a first end of the shaft **116** at a pivot **124**. The pivot **124** can allow the housing portion **104** and mirror **108** to be pivoted in one or more directions (e.g., up, down, right, left, and/or in any other direction). For example, the pivot **124** can include a ball joint, one or more hinges, or otherwise.

(24) In some embodiments, the shaft **116** can be pivotably coupled to the stand **120** at a second end of the shaft **116** at a pivot **128**. The pivot **128** can allow the shaft **116** to be pivoted in one or more directions (e.g., up, down, right, left, and/or in any other direction). The shaft **116** can be pivoted to transition the mirror assembly **100** to a stowed position as shown in FIGS. **9** and **10**. The shaft **116** can be pivoted to rest in a recess **132** of the stand **120**.

(25) The mirror **108** can include a generally flat, curved, or generally spherical surface, which can be convex or concave. The radius of curvature can depend on the desired optical power. In some embodiments, the radius of curvature can be at least about 15 inches and/or less than or equal to about 30 inches. The focal length can be half of the radius of curvature. For example, the focal length can be at least about 7.5 inches and/or less than or equal to about 15 inches. In some embodiments, the radius of curvature can be at least about 18 inches and/or less than or equal to about 24 inches. In some embodiments, the mirror **108** can include a radius of curvature of about 20 inches and a focal length of about 10 inches. In some embodiments, the mirror **108** is aspherical, which can facilitate customization of the focal points.

(26) In some embodiments, the radius of curvature of the mirror **108** is controlled such that the magnification (optical power) of the object is at least about 2 times larger and/or less than or equal to about 7 times larger. In certain embodiments, the magnification of the object is about 5 times larger. In some embodiments, the mirror can have a radius of curvature of about 19 inches and/or about 7 times magnification. In some embodiments, the mirror can have a radius of curvature of about 24 inches and/or about 5 times magnification.

(27) As shown, the portion of the mirror assembly **100** that faces the user is the mirror **108**, which can have a generally circular shape. In some embodiments, the mirror **108** can have an overall shape that is generally elliptical, generally square, generally rectangular, or any other shape. In some embodiments, the mirror **108** can have a diameter of at least about 8 inches and/or less than or equal to about 12 inches. In some embodiments, the mirror **108** can have a diameter of about 8 inches. In certain embodiments, the mirror **108** can have a diameter of at least about 12 inches and/or less than or equal to about 16 inches. In some embodiments, the mirror **108** can include a thickness of at least about 2 mm and/or less than or equal to about 3 mm. In some embodiments, the thickness is less than or equal to about two millimeters and/or greater than or equal to about three millimeters, depending on the desired properties of the mirror **108** (e.g., reduced weight or greater strength).

(28) The mirror **108** can be highly reflective (e.g., has at least about 90% reflectivity). In some embodiments, the mirror **108** has greater than about 70% reflectivity and/or less than or equal to about 90% reflectivity. In other embodiments, the mirror **108** has at least about 80% reflectivity and/or less than or equal to about 100% reflectivity. In certain embodiments, the mirror **108** has about 87% reflectivity. The mirror **108** can be cut out or ground off from a larger mirror blank so that mirror edge distortions are diminished or eliminated. One or more filters can be provided on the mirror to adjust one or more parameters of the reflected light. In some embodiments, the filter comprises a film and/or a coating that absorbs or enhances the reflection of certain bandwidths of electromagnetic energy. In some embodiments, one or more color adjusting filters, such as a Makrolon filter, can be applied to the mirror to attenuate desired wavelengths of light in the visible spectrum.

(29) The mirror **108** can be optical grade and/or comprise glass. For example, the mirror **108** can include ultra clear glass. Alternatively, the mirror **108** can include other transparent or translucent materials, such as crystal, plastic, nylon, acrylic, or other suitable materials. The mirror **108** can also include a coating or backing **110** that has high reflectivity and low transmissivity, such as a coating or backing **110** that includes a metal such as aluminum or silver. The coating **110** can be on a rear surface of the mirror **108** (e.g., the surface facing away from a user). In some embodiments, the coating **110** can impart a slightly colored tone, such as a slightly bluish tone to the mirror. In some embodiments, an aluminum coating can prevent rust formation and provide an even color tone. The mirror **108** can be manufactured using molding, machining, grinding, polishing, or other techniques.
(30) The mirror assembly **100** can include one or more light sources **136** configured to transmit light. In some embodiments, the light source **136** can surround an outer perimeter of the mirror **108**, for example, the light source **136** can be a ring as shown. Light can be evenly or generally uniformly distributed around the entire perimeter of the mirror **108** without any optically significant breaks or interruptions in the light emitting perimeter (such as may otherwise occur if a proximity sensor is positioned along or near the perimeter of the mirror assembly **100**). In some embodiments, one or more light sources **136** can be positioned on or around a user facing side of the mirror assembly **100**. In some embodiments, one or more light sources **136** can be positioned on a side of the housing **104**. Various light sources **136** can be used. For example, the light sources **30** can include light emitting diodes (LEDs), fluorescent light sources, incandescent light sources, halogen light sources, or otherwise.
(31) FIG. **11** illustrates the mirror assembly **100** with the first mirror **108** removed for illustrative purposes. FIG. **12** illustrates the mirror assembly **100** with the housing **104** removed for illustrative purposes. As shown in FIGS. **11** and **12**, a second mirror **121** can be positioned behind and/or coupled directly or indirectly to the first mirror **108**. In some embodiments, the first mirror **108** and the second mirror **121** can both be curved or non-planar as shown, such as with corresponding curvatures that permit the first mirror **108** and second mirrors **121** to fit closely together without any functionally significant gap between them. A rear surface of the first mirror **108** can be coupled with a front surface of the second mirror **121**. In some embodiments, the first mirror **108** and the second mirror **121** can be placed adjacent to each other or in contact with each other when the mirror assembly **100** is assembled, such as with a mechanical clip or bracket, or with optical adhesive **114**. In some embodiments, the optical adhesive **114** can be made of a material that has the same or substantially the same index of refraction as the transparent material **109** of the first mirror **108** so as to resist refraction of light at the boundary between the transparent material **109** and the optical adhesive **114**. The optical adhesive **114** can be used to couple a front surface **111** of the second mirror **121** to a rear surface **110** of the first mirror **108** and/or to fill any gap or avoid that may otherwise occur between or near them.
(32) In some embodiments, the second mirror **121** can be smaller than the first mirror **108**. In some embodiments, the first mirror **108** and the second mirror **121** can be the same shape but of different sizes, for example, the first mirror **108** can be a larger circular shape and the second mirror **121** can be a smaller circular shape. While circular mirrors **108**, **121** are depicted, any shape mirrors can be used, for example, square, oval, and rectangular. The first mirror **108** and the second mirror **121** can have essentially the same reflectance. The first mirror **108** can have very low transmissivity (e.g., less than or equal to about 15%) to visible light and infrared light, and/or it can have essentially no observable transmissivity by a user to visible light, while the second mirror **121** can have a high transmissivity to infrared light (e.g., at least about 75% transmissivity for infrared light) and it can have very low transmissivity to visible light (e.g., at least about 80% reflectivity). The transmissivity and reflectivity of the first mirror **108** and the second mirror **121** can be similar to each other (e.g., within about 15%) for visible light but substantially different from each other for infrared right (e.g., differing by at least about 50%).
(33) In some embodiments, the second mirror **121** can be positioned behind a window **122** of the first mirror **108**, for example, as shown in FIG. **13**. The window **122** does not extend through a thickness of the first mirror **108**. The window **122** can be a portion of the first mirror **108** that does not have the coating **110** or backing covering it. The window **122** can permit light to be transmitted through the transparent material **109** of the mirror **108** in the region of the window **122**). The window **122** can be highly transmissive (e.g., essentially transparent). When positioned in the complete mirror assembly **100**, the window **122** and the components behind it can be almost invisible to a user during use, thus hiding the appearance of a sensor assembly, discussed in more detail below.
(34) The second mirror **121** can have a layer, backing, or coating **111** on a front surface of a transparent material **113**. The coating **111** can be the surface coupled to the coating **110** of the first mirror **108**. The coating **111** can comprise a series or stack of different and/or alternating materials, such as two or more metal oxide, silicon oxide, and/or dielectric layers, to produce a reflective optical interference effect. The color of the coating **111** can be adjusted by choosing suitable materials and/or thicknesses of the materials in the stack to match the color or appearance of the coating **110** on the rear surface of the first mirror **108**. Substantially matching the color or appearance of the coating **110** on the rear surface of the first mirror **108** and the coating **111** on the front surface of the second mirror **121** can assist in hiding (e.g., making substantially invisible) the window **122** and the components behind it from the user during use. In some embodiments the mirror **121** can include a transparent material such as

glass and one or more layers. In some embodiments, the mirror **121** comprises one or more optical interference coatings or layers such as dielectric glass coatings or layers (e.g., glass with dielectric layers). In some embodiments, the mirror **121** can include an optical interference structure comprising one or more extremely thin layers (e.g., with a thickness that is less than a single wavelength of visible light) that are configured to produce an optical effect such as a pre-determined reflectance and/or transmissivity of visible light at particular wavelengths or groupings of wavelengths. For example, in some embodiments, the optical interference structure can comprise a series of layers comprising at least one metallic layer and at least one dielectric layer, such as one or more alternating layers of Titanium dioxide (TiO2) and Silicon Dioxide (SiO2). The one or more alternating layers can be applied by chemical or vapor deposition.
(35) When the rear surface of the first mirror **108** is coupled to the front surface of the second mirror **121** the coating **110** on the rear surface of the first mirror **108** and the coating **111** on the front surface of the second mirror **121** can function optically to a user's perception as one coating. For example, as shown in FIG. **14**, the assembly **126** of the first mirror **108** and the second mirror **121** can comprise a first glass layer, alternating layers of TiO2 and SiO2, metal (e.g., aluminum), and a second glass layer. The assembly **126** can include a layer of SiO2 on each side of a metal layer. A protective or opaque paint, coating, or layer can be included directly on the glass or on a metal layer and/or can be layered on top of a SiO2 layer or any other layer or structure of the assembly **126**. As used in this specification, the term “glass” is used in accordance with its ordinary meaning, which includes a silica-based hard, brittle, transparent material, typically formed in thin layers or panes. With each occurrence of “glass,” it is also contemplated that non-glass substances or alternatives that are structured in a manner similar to glass or that perform in substantially the same way as glass can be used, including one or more transparent coated or uncoated polymer sheet materials such as those comprising polycarbonate or acrylic. As with all disclosure in this application, any layer or substrate or combination of layers or substrates that are disclosed in connection with this embodiment can be replaced with or combined with any other layer or substrate or combination of layers or substrates that are disclosed elsewhere herein, or omitted.
(36) In some embodiments, the mirror assembly **100** in one region can have a first layer **109**, a second layer **110**, a third layer **111**, and a fourth layer **113**. The material(s) of which each layer **109**, **110**, **111**, **113** are made can comprise glass, metal (e.g., aluminum, silver, titanium dioxide, etc.), and a dielectric material (e.g., SiO2, etc.), and/or any other suitable material. In some embodiments, the mirror **108** can be assembled using a sputtering process, a vaporizing process, a deposition process, and/or any other suitable process. In some embodiments, one or more of the layers can be applied using adhesive, thermal bonding, sonic welding, or in any other suitable way. In some embodiments, the first layer **109** comprises glass. This can help prevent the mirror **108** from becoming scratched because glass is highly scratch resistant, much more scratch resistant than many other materials. In some embodiments, the second layer **110**, the layer disposed between the first layer **109** and the third layer **111**, comprises a metal (e.g., aluminum). In some embodiments, the third layer **111** comprises a metallic or dielectric material (e.g., TiO2, SiO2). In some embodiments, the fourth layer **113** comprises glass. One or more of the layers **109**, **110**, **111** can be used in the mirror **121**. One or more of the layers **109**, **110**, **111**, **113** can be omitted from the assembly **126**, rearranged or assembled in a different order, and/or supplemented by other layers or materials. For example, in some embodiments, a dark coating such as black paint and/or a copper layer is disposed on at least a portion of a metal layer of the mirror **108**. Any layer or structure disclosed in this specification is not required to be homogeneous, but can include multiple sub-layers or sub-structures as appropriate.
(37) In some embodiments, an optical interference structure in the mirror **108** can include a combination of layers such as the following:
Example 1
(38) TABLE-US-00001 Optical Physical Layer Material Thickness Thickness (nm) Substrate protective paint 9 Al 0.48 120 8 SiO2 1.0768 72.93 7 TiO2 1.2789 53.33 6 SiO2 1.2948 87.7 5 TiO2 1.3201 55.04 4 SiO2 2.3774 161.03 3 TiO2 1.3064 54.47 2 SiO2 2.1806 147.69 1 TiO2 2.0414 85.12 Medium glass Total Thickness 13.3564 837.31
Example 2
(39) TABLE-US-00002 Optical Physical Layer Material Thickness Thickness (nm) Substrate protective paint 10 SiO2 0.7382 50 9 Al 0.48 120 8 SiO2 1.0768 72.93 7 TiO2 1.2789 53.33 6 SiO2 1.2948 87.7 5 TiO2 1.3201 55.04 4 SiO2 2.3774 161.03 3 TiO2 1.3064 54.47 2 SiO2 2.1806 147.69 1 TiO2 2.0414 85.12 Medium glass Total Thickness 14.0946 887.31
(40) Of course, many other types and orderings of layers can be used instead of or in addition to those provided in this example. In some embodiments, a reflectance of a mirror with an optical interference structure on a glass substrate can be provided as shown in FIG. **17**.
(41) For example, as shown in FIG. **17**, for a large majority of the wavelengths of visible light, the reflectance can be at least about 95% or at least about 93% or at least about 90%. Of course, any other suitable component or series of components can be configured to provide many other or different values in a useful reflectance profile in accordance with the principles and structures disclosed in this specification.

(42) The relative sizes of the coating layers **110**, **111** as illustrated in FIG. **14** is exaggerated as compared to the transparent material **109**, **113**, for case of illustration. The coating layers **110**, **111** can be significantly thinner than as shown as compared to the transparent materials **109**, **113**.
(43) As shown in FIG. **14**, the first mirror **108** can be configured to permit visible light to pass through the front transparent material **109**, reflect off of the rear reflective coating or backing **110**, pass through the front transparent material **109** again, and then enter the eye of a user, forming a reflected image of the user. The combination of the first mirror **108** and the second mirror **121** can be configured to permit visible light impinging on the region of the first mirror **108** in front of the window **122** to pass through the front transparent material **109**, enter the window **122** (which can be filled with optical adhesive **114**), pass through the optical adhesive layer **114**, reflect off of the front reflective interference coating **111** of the second mirror **121**, pass through the optical adhesive layer **114** and the window **122** again, pass through the transparent material **109** again, and then enter the eye of a user, forming a reflected image of the user. The mirror assembly **100** can be configured to effectively hide the window **122** and the components behind it from the user by permitting visible light to be reflected in the region of the window **122** in essentially the same manner and at about the same reflectivity as in the regions of the first mirror **108** that are adjacent to and/or surrounding the window **122**.
(44) Simultaneously, the transmitter **141** of the sensor **140** can emit sensing energy, such as electromagnetic energy in the infrared range, through the transparent material **113** of the second mirror **121**, through the coating **111**, through the optical adhesive, through the window **112**, and through the transparent material **109** of the first mirror **109** without substantial refraction, reflection, deviation, attenuation, and/or modification. As shown, the emitted sensing energy can reflect off of the user and then return through the transparent material **109** of the first mirror **108**, window **122**, optical adhesive **114**, coating **111**, and transparent material **113** of the second mirror **121**, and then proceed into the receiver **142** of the sensor **140**, again without substantial refraction, reflection, deviation, attenuation, and/or modification. The sensing energy is not reflected in a significant way by the coating **111** because the interference effects of the coating **111** are calculated to produce reflection for only certain wavelengths of light within the visible range. For example, the coating **111** can be configured to reflect impinging visible light in the same way as the coating **110** of the first mirror **108** so that the reflected visible light in the region of the window **122** is the same as in the other regions of the first mirror **108**.
(45) FIG. **15** illustrates a front view of the mirror assembly **100** with the mirror **108** removed for illustrative purposes and FIG. **16** illustrates a rear view of the mirror assembly **100** with a rear portion of the housing **104** removed for illustrative purposes. The mirror assembly **100** can include a sensor assembly **140**. The sensor assembly **140** can be positioned between a rear surface of the mirror **108** and a rear surface of the housing **104**. The sensor assembly **140** can be enclosed by the mirror **108** and the housing **104**. The sensor assembly **140** can be hidden such that it is not visible by a user. In some embodiments, the sensor assembly **140** can be positioned generally central within the housing **104**. The sensor assembly **140** can include a sensor **144**. In some embodiments, the sensor **144** can be a proximity sensor or a reflective-type sensor. For example, the sensor **144** can be triggered when an object (e.g., a body part) is moved into, and/or produces movement within, a sensing region.
(46) The sensor assembly **140** can include a power source **148** (e.g., a battery, a rechargeable battery, or a cord to be plug into an electrical outlet). The power source **148** can be positioned within the housing **104**. The power source **148** can deliver power to the light source **136**. In some embodiments, the mirror assembly **100** can include one or more weights **145** (labeled in FIG. **8**) disposed in the stand **120** of the mirror assembly **100** to counteract the weight of the sensor assembly **140**.
(47) The sensor assembly **140** can include a transmitter **141** and a receiver **142**. The transmitter **141** can be an emitting portion (e.g., electromagnetic energy such as infrared light), and the receiver **142** can be a receiving portion (e.g., electromagnetic energy such as infrared light). The beam of light emitting from the transmitter **141** can define a sensing region. In certain variants, the transmitter **141** can emit other types of energy, such as sound waves, radio waves, or any other signals. The transmitter **141** and receiver **142** can be integrated into the same sensor or configured as separate components. The transmitter **141** and the receiver **142** can be positioned behind a rear surface of the second mirror **121**. The transmitter **141** and the receiver **142** can be aligned with and/or positioned behind the window **122** of the first mirror **108**.
(48) In some embodiments, the mirror assembly **104** can include an LED light **143**. The LED light **143** can be positioned behind a rear surface of the second mirror **121**. The LED light **143** can be aligned with and/or positioned behind the window **122** of the first mirror **108**. The LED light **143** can be configured to emit a light in order to alert a user to a predetermined type of information. In some embodiments, the light can be emitted to indicate in one or more different ways (e.g., changing color, flashing, emitting in a steady state, etc.) that the battery of the mirror assembly **100** is charging, finished charging, and/or nearing depletion. In some embodiments, the light can be emitted to indicate that a user's presence has been identified.
(49) In some embodiments, the sensor assembly **140** can detect an object within a sensing region. In certain embodiments, the sensing region can have a range from at least about 0 degrees to less than or equal to about 45

degrees downward relative to an axis extending from the sensor assembly **28**, and/or relative to a line extending generally perpendicular to a front surface of the sensor assembly, and/or relative to a line extending generally perpendicular to the front face of the mirror and generally outwardly toward the user from the top of the mirror assembly. In certain embodiments, the sensing region can have a range from at least about 0 degrees to less than or equal to about 25 degrees downward relative to any of these axes or lines. In certain embodiments, the sensing region can have a range from at least about 0 degrees to less than or equal to about 15 degrees downward relative to any of these axes or lines.

(50) If the receiving portion detects reflected energy (e.g., above a threshold level) from an object within the beam of light emitted from the transmitter **141**, the sensor assembly **140** sends a signal to a controller to activate the light source **136**. Once the light source **136** activates, the light source **136** can remain activated so long as the sensor assembly **140** detects an object in a sensing region. Alternatively, the light source **136** can remain activated for a pre-determined period of time. For example, activating the light source **136** can initialize a timer. If the sensor assembly **140** does not detect an object before the timer runs out, then the light source **136** is deactivated. If the sensor assembly **140** detects an object before the timer runs out, then the controller can reinitialize the timer, either immediately or after the time runs out.

(51) Conditional language, such as "can," "could," "might," or "may," unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments.

(52) The terms "comprising," "including," "having," and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term "or" is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term "or" means one, some, or all of the elements in the list.

(53) The ranges disclosed herein also encompass any and all overlap, sub-ranges, and combinations thereof. Language such as "up to," "at least," "greater than," "less than," "between" and the like includes the number recited. Numbers preceded by a term such as "about" or "approximately" include the recited numbers. For example, "about 5 mm" includes "5 mm."

(54) The term "horizontal" as used herein is defined as a plane parallel to the ground or floor on which the component or device is positioned in normal use, either directly or indirectly. The term "vertical" refers to a direction perpendicular to the horizontal as just defined. Terms such as "above," "below," "bottom," "top," "side," "higher," "lower," "upper," "over," and "under," are defined with respect to the horizontal plane.

(55) Although certain embodiments and examples have been described herein, it will be understood by those skilled in the art that many aspects of the devices shown and described in the present disclosure may be differently combined and/or modified to form still further embodiments or acceptable examples. All such modifications and variations are intended to be included herein within the scope of this disclosure. A wide variety of designs and approaches are contemplated. No feature, structure, or step disclosed herein is essential or indispensable.

(56) For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. It is to be understood that not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those skilled in the art will recognize that the disclosure may be embodied or carried out in a manner that achieves one advantage or a group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

(57) Moreover, while illustrative embodiments have been described herein, the scope of any and all embodiments having equivalent elements, modifications, omissions, combinations (e.g., of aspects across various embodiments), adaptations and/or alterations as would be appreciated by those in the art based on the present disclosure. The limitations in the claims are to be interpreted broadly based on the language employed in the claims and not limited to the examples described in the present specification or during the prosecution of the application, which examples are to be construed as non-exclusive. Further, the actions of the disclosed processes and methods may be modified in any manner, including by reordering actions and/or inserting additional actions and/or deleting actions. It is intended, therefore, that the specification and examples be considered as illustrative only, with a true scope and spirit being indicated by the claims and their full scope of equivalents.

Claims

1. A mirror assembly comprising: a first mirror with a front surface and a rear surface, the front surface visible by a user; a second mirror having a front surface and a rear surface, the front surface of the second mirror coupled to the rear surface of the first mirror; a light source disposed on or at least partially around the first mirror; and a sensor assembly positioned behind a rear surface of the second mirror, the sensor assembly comprising: a transmitter

positioned behind the rear surface of the second mirror; and a receiver positioned behind the rear surface of the second mirror; wherein the light source is configured to be activated by the sensor assembly.

2. The mirror assembly of claim 1, wherein the sensor assembly comprises a proximity sensor.

3. The mirror assembly of claim 1, wherein the transmitter and the receiver are positioned behind a window of the first mirror.

4. The mirror assembly of claim 3, further comprising an LED light positioned behind the window of the first mirror, wherein the LED light can be configured to emit a light when the mirror assembly is charging.

5. The mirror assembly of claim 3, wherein the window is substantially invisible to the user.

6. The mirror assembly of claim 1, wherein the second mirror is smaller than the first mirror.

7. The mirror assembly of claim 1, wherein the first mirror and the second mirror have the same curvature.

8. The mirror assembly of claim 1, wherein the second mirror is coupled to the rear surface of the first mirror via an adhesive.

9. The mirror assembly of claim 8, wherein the index of refraction of the adhesive is substantially the same as the index of refraction of a transparent material of the first mirror.

10. The mirror assembly of claim 1, wherein the first mirror has a coating on the rear surface and the second mirror has a coating on the front surface.

11. The mirror assembly of claim 10, wherein the coating of the first mirror is aluminum and the coating of the second mirror is an optical interference coating.

12. The mirror assembly of claim 1, further comprising a power source positioned within a housing of the mirror assembly.

13. The mirror assembly of claim 1, further comprising a shaft coupled at a first end to a housing of the mirror assembly and at a second end to a stand.

14. The mirror assembly of claim 13, wherein the shaft is rotatably coupled at the first end to the housing and the shaft is rotatably coupled at the second end to the stand.

15. A mirror assembly comprising: a first mirror with a front surface and a rear surface, the front surface visible by a user; a second mirror having a front surface and a rear surface, the front surface of the second mirror coupled to the rear surface of the first mirror; and a sensor assembly positioned behind the rear surface of the second mirror, the sensor assembly comprising: a transmitter configured to emit light; and a receiver configured to receive light; wherein the sensor assembly is configured to detect a user's presence.

16. The mirror assembly of claim 15, wherein the sensor assembly comprises a proximity sensor.

17. The mirror assembly of claim 15, further comprising a light source, wherein when a user's presence is detected the light source is activated.

18. The mirror assembly of claim 15, wherein the transmitter and the receiver are positioned behind a window of the first mirror.

19. The mirror assembly of claim 15, wherein the second mirror is smaller than the first mirror.

20. The mirror assembly of claim 15, wherein the first mirror and the second mirror have the same curvature.

21. The mirror assembly of claim 15, wherein the second mirror is coupled to the rear surface of the first mirror via an adhesive.

22. The mirror assembly of claim 15, wherein the first mirror has a coating on the rear surface and the second mirror has a coating on the front surface.

23. The mirror assembly of claim 22, wherein the coating of the first mirror is aluminum and the coating of the second mirror comprises a dielectric layer.
